Implementing Hybrid Quantum Neural Networks for Accurate Wildfire Detection

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***Abstract*—This study presents a pioneering approach to wildfire prediction through a Hybrid Quantum-Classical Neural Network (HQCNN), addressing the critical need for more accurate wildfire detection amid rising wildfire frequency and intensity. Traditional machine learning techniques, while effective, often encounter limitations like overfitting due to high computational demands. By incorporating a variational quantum circuit into a hybrid framework, the HQCNN leverages quantum mechanics to enhance feature extraction, achieving superior computational efficiency and predictive performance. Our model attained a testing accuracy of 95.50%, significantly outperforming the 88.43% achieved by a purely classical model. This marked improvement in accuracy, alongside reduced training time, highlights the transformative potential of quantum-enhanced models for environmental applications. Our results underscore the strategic advantage of quantum integration in machine learning tasks, setting a new benchmark for wildfire prediction and demonstrating the broader implications of quantum approaches for complex classification challenges.**

***Index Terms*—Wildfires, Quantum Computing, Machine Learning, Variational Quantum Circuit**

I. Introduction

The recent surge in wildfires has wreaked havoc on natural habitats and driven unparalleled biodiversity loss. In 2021, Russia witnessed the destruction of 5.4 million hectares of forest due to fires [1]. The following year, the California Mosquito Fire in the USA destroyed 78 structures in surrounding communities alongside thousands of hectares of forest, causing an estimated 3.2 billion USD in damage [1]. By 2023, Canada faced its worst year yet, with over 9 million hectares of land scorched [1].

While Machine Learning (ML) techniques have seen some success in developing effective early warning systems, they are hindered by their high computational costs [2] and complexity [3], especially given the growing size and complexity of datasets. The training of deep learning algorithms, a subset of ML used in image classification tasks, can be classified as an NP-complete problem with no computationally efficient solution [4]. Therefore, these architectures often require extensive training periods, ranging from several hours to multiple months on current computers, while also leading to overfitting issues [5].

However, quantum computers have emerged as a promising solution in this area, offering the potential to develop effective detection systems for wildfires [5]. By exploiting quantum features, such as entanglement and superposition, quantum computers store and encode data in a way which cannot be efficiently simulated on a classical computer [6]. Therefore, quantum algorithms can lead to several benefits over classical counterparts, offering exponential computational boosts [6] and the ability to simultaneously process more inputs with fewer operations.

In ML, quantum algorithms provide significant speedups in data handling by efficiently managing high-dimensional vectors, essential for large dataset processing [7], [8]. Furthermore, quantum random access memory (qRAM) enables a large amount of classical data to be represented succinctly by encoding a vector with N components in just $\log N$ qubits [8]. In principle, quantum computers can provide improvements for various ML subroutines such as data fitting [9], support vector machines [10] and Monte Carlo methods [11] [12]. However, the current development phase — known as the *Noisy Intermediate Scale Quantum (NISQ) era* — faces challenges due to the limited and noisy qubits on quantum hardware, complicating the realization of standalone, fully functional quantum algorithms [7].

Due to the advantages of quantum computing such as using significantly less features than classical neural networks [13], hybrid quantum-classical neural networks appear to be a perfect compromise. Classical layers extract large-scale features, maintaining the locality and non-linearity of convolutional layers [3]. This drastically reduces the number of training parameters as the same set of weights (filter) is used across different regions of the input [3], [14]. Quantum layers are then used to detect abstract features within the constraints of NISQ technology [5], increasing the accuracy of the model.

This study is dedicated to the study of Variational Quantum Algorithms (VQAs) which have been recognized as the prominent analogue to quantum layers [3]. VQAs run parametrized quantum circuits on the quantum computer, and then outsource the parameter optimization to a classical optimizer [15]. Due to their versatility, VQAs have applications in quantum chemistry [16], quantum optimization [17], quantum machine learning [12], [18]–[20] and other fields.

Since classical algorithms struggle to quickly and accurately analyze substantial datasets while also examining complex, non-linear features [21], this paper explores how powerful hybrid quantum neural network systems can be implemented to identify whether an area is at risk to a wildfire. This could increase the efficacy of early detection systems that are crucial for decision-makers in designing mitigation and extinguishing strategies [22] [23].

The purpose of this research is not only to advance machine learning methodologies for wildfire detection, but also to extend Quantum-Classical ML solutions to broader environmental monitoring applications. This approach, exemplified by the potential combination of Quantum Federated Learning (QFL) and Quantum Support Vector Machines (QSVM) across distributed quantum processors, reflects a transformative leap in data processing and analysis. As quantum and classical computing techniques continue to merge, this work presents a potential paradigm shift in large-scale environmental applications where the analysis of complex, high-dimensional data is essential.

II. Background Information

A. Classical Artificial Neural Networks

The feed-forward neural network is a fundamental aspect of modern neural networks [24]. It is composed of multiple *layers*, each of which maps input vectors of n_0 elements to output vectors of n_1 elements. A layer is composed of an affine operation followed by a non-linear function ψ,

$$L_i : x_i \longrightarrow y_i = \psi(Wx_i + b) \tag{1}$$

where L_i is the specified layer, W is the weight vector, x_i is the input feature vector, y_i is the output feature vector and b is the bias. The output from one layer is then fed as input into the next layer, meaning the network can be represented as a concatenation of layers. Summarizing all model parameters by the parameter set θ, an N-layer neural network model is given by

$$y = f_\theta(x) = L_n \circ \cdots \circ L_1(x) \tag{2}$$

and maps an input x to a final output y.

B. Fundamental Quantum Properties

A variational quantum circuit uses quantum properties such as superposition and entanglement to encode and process data [3].

In the Z-basis, the spin-up and spin-down states, $|0\rangle$ and $|1\rangle$, can be expressed in matrix form as follows:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{3}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{4}$$

Quantum superposition refers to a system that is in multiple states until a measurement is taken. The superposition of a given state $|\psi\rangle$ in the z-basis is given by:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{5}$$

where coefficients α and β are the corresponding probability amplitudes of the orthogonal z-basis vectors $|0\rangle, |1\rangle$.

Quantum entanglement is the principle which states that once a pair of systems become entangled, it is impossible to measure the state of one system without knowing the state of the other. A fundamental representation of entanglement is the Bell State, where $|0\rangle_0, |1\rangle_0$ are the basis vectors of the first system and $|0\rangle_1, |1\rangle_1$ are basis vectors of the second system, can be written as:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle_0|0\rangle_1 + |1\rangle_0|1\rangle_1\right) \tag{6}$$

C. Variational Quantum Circuits

In a hybrid quantum classical neural network, the purpose of a variational quantum circuit (VQC) to encode and manipulate data from classical neural networks [3]. To do this, VQCs utilise quantum gates that perform operations on systems defined by quantum properties.

Generally, a VQC is composed of three layers - embedding, variational and measurement - each using a sequence of quantum gates.

In order to insert classical data into a quantum network, we need to embed a real vector x into a quantum state $|x\rangle$. First, a classical parameter is used to map x to the quantum Hilbert space. Then, an embedding function E parameterized by x is applied to a reference state (e.g., the vacuum or ground state):

$$\epsilon : x_i \mapsto |x_i\rangle = E(x)|0_i\rangle \tag{7}$$

Common embedding functions include single-qubit rotations or single-mode displacements.

Drawing a parallel to the classical scenario, the variational layer is composed of a unitary operation U that is implementable through a low-depth variational circuit [19]. This variational layer carries out qubit entanglement and rotation, transforming them from an initial state to a target state that enables complex, non-linear information mapping.

This circuit acts on the input state $|x\rangle$ of n_q quantum subsystems (such as qubits) and generates the output state $|y\rangle$.

$$L : |\psi(x)\rangle \rightarrow |\psi(y)\rangle = U(w)|\psi(x)\rangle \tag{8}$$

$$|\psi(x)\rangle = O \bigotimes_{k=1}^{n} |x\rangle_n \tag{9}$$

where $|\psi(x)\rangle$ is the n qubit input state and w is an array of classical variational parameters. Generally, variational circuits consist of single-qubit rotations followed by a series of entangling gates [19].

A variational quantum circuit with depth q is composed of a sequence of quantum layers, each represented by unitary operators with different parameters. The circuit can be expressed as:

$$Q = L_q \circ \cdots \circ L_2 \circ L_1. \tag{10}$$

The extraction of a classical output vector y can be obtained by taking a measurement of previously defined n_q observables $\hat{y} = [\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_{n_q}]$ [19]. This can be defined in a measurement layer as

$$M : |x\rangle \rightarrow y = \langle x|\hat{y}|x\rangle \tag{11}$$

Overall, the global variational quantum circuit V can be expressed as

$$V = M \circ Q \circ E \tag{12}$$

where E is the embedding layer, Q is the variational circuit, and M is the measurement layer in the Z-basis.

In essence, the concept of a Quantum-Classical Convolutional Neural Network is to utilize a parametric quantum circuit to create the feature map within the convolutional layer. Consequently, the output of this feature map is derived from a correlational measurement performed on the output quantum state of the parametric quantum circuit.

D. Related Works

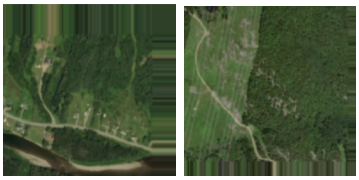
Liu et al. [3] introduced the framework of hybrid quantum-classical convolutional neural networks (HQCNNs). The architecture featured a quantum convolutional layer followed by classical pooling and fully connected layers. In the quantum convolutional layer, a quantum filer taking sections of size mxn is implemented, mapping them into quantum states of $N = mn$ qubits. These states are evolved using a VQC, and the output is fed into classical pooling layers and a fully connected layer to to further introduce non-linearity [3]. After training on the TETRIS dataset, n-layerd HQCNNs were shown to outperform corresponding classical convolutional neural networks in terms of accuracy [3].

Killoran et al. [12] investigated the use of fully connected classical layers followed by a continuous variable (CV) architecture. The CV model utilizes the wave-like characteristics of nature: instead of encoding quantum information in qubits, it relies on the quantum states of fields, such as the electromagnetic field [12]. A CV quantum neural network is a sequence of layers, with each layer containing every gate from the universal gate set - an interferometer, local squeeze gates, a second interferometer, local displacements, and local non-Gaussian gates [12]. Paired with classical fully connected layers, the CV model outputs resembling very closely to noiseless ground truth curves for the task of curve fitting [12]. Choe [26] builds an image classification model that use classical feed-forward layers for initial data processing and the introduction of non-linearity followed by this CV quantum neural network [12], reaching an accuracy of 99.90% on the MNIST dataset using an eight-class classifier [26].

III. Materials and Methods

A. Description of Dataset

The image dataset used in this study was produced and published publicly on Kaggle by Abdelghani Aaba [28]. The source data was obtained from Canada's website for forest fires, and, using Longitude and Latitude coordinates for each wildfire spot (greater than 0.01 acres burned) found on the dataset above, satellite images of those areas were extracted using MapBox API. Overall, there are 22710 wildfire images and 20140 non-wildfire images.



(a) **(b)**

Fig. 1: Example Training Images - (a) is a wildfire prone area whereas (b) is a non-wildfire prone area.

B. Prepossessing of Dataset

The data was divided into train (80%), test (10%) and validation (10%) classes. For the purposes of this study, 1000 images were randomly sampled for training, and 200 images for validation. Images was resized to 224 x 224 pixels as this is the standard used in ResNet Models. The image tensor was normalized by scaling the pixel values to have a mean of $[0.485, 0.456, 0.406]$ and a standard deviation of $[0.229, 0.224, 0.225]$ for each of the three color channels (red, green, and blue) respectively, matching the normalization values used for the ImageNet dataset. This is necessary because pre-trained models are typically trained on ImageNet, so using these normalization values helps ensure that the input images are on the same scale as the data the model was originally trained on.

C. Variational Quantum Circuit

The VQC used in this paper is similar to architectures developed by Mari et al. [19]. That is to say we implement a hybrid classical to quantum transfer learning model in this section of the paper.

1) Embedding Layer: First, Hadamaard (H) Gates are implemented to bring 4 qubits into superposition states. An H Gate can be represented as a matrix, shown below:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{13}$$

A standard kernel function is used to to map each element of the array into a single-qubit state. For the purposes of this study, the real input vector is mapped to the Quantum Hilbert Space by function θ, which is defined as

$$\theta(k) = \tanh(I_{i,j}) \cdot \pi/2 \tag{14}$$

where the term $I_{i,j}$ denotes the intensity of the pixel located at coordinates (i, j) within the convoluted input image. The variable k specifies a particular qubit number. For the four-qubit variational circuit utilized in this study, k takes values in {1, 2, 3, 4}, and the parameters for the pixel at position (i, j) are represented by $\theta(k)$.

A rotation-y (RY) Gate is then applied, rotating the qubit by a given number of radians θ, encoding the qubit with the classical input vectors. It can be expressed in matrix form as:

$$R_Y(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{15}$$

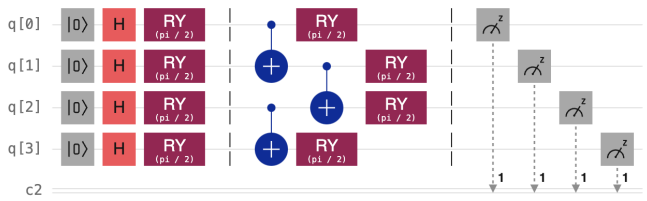


Fig. 2: IBM Quantum Composer Representation of proposed VQC at depth 1

Overall, the embedding layer of this circuit can be defined as

$$\epsilon(x) = R_y(\theta_{i,j}^{(k)})H \tag{16}$$

2) Variational Layer: The variational layer consists of alternating CNOT and RY Gates. Each CNOT Gate operates by examining the control qubit: if the control qubit is in the state $|1\rangle$, it will flip the target qubit from $|0\rangle$ to $|1\rangle$. The matrix representation is below:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{17}$$

Therefore, the underlying unitary operation, T, of the variational layer can be defined as

$$T = C_{1,2}C_{3,4}C_{2,} \tag{18}$$

where the first C subscript represents the control qubit and the second subscript represents the target qubit for CNOT gates.

3) Measurement Layer: The final layer measures the quantum states of the 4 qubits using a Pauli Z measurement.

The circuit is clearly outlined in Figure 2.

D. Classical Convolutional Neural Network

Convolutional neural networks (CNNs) are used field of image pattern recognition. By encoding image-specific features directly into the architecture, CNNs reduce the number of parameters needed to configure the model. Compared to artificial neural networks (ANNs), CNNs offer reductions in computational complexity and are less prone to overfitting [3].

E. Hybrid Quantum-Classical Neural Network

For the base model of the hybrid neural network, ResNet18—a classical, sequential CNN model—was selected due to its proven effectiveness in image classification tasks in recent years [27]. The model comprises multiple convolutional and pooling layers, enabling it to analyze large datasets effectively [27]. Using PyTorch, a pretrained ResNet18 model was loaded for further processing.

To incorporate the quantum circuit, 512 features were extracted from the ResNet18 average pooling layer and used as inputs for the quantum layer. The final layer of the ResNet18 architecture was removed and replaced by a *dressed quantum net* [19] implemented in PyTorch. First, the dressed quantum net reduced the feature dimensionality and applied the previously defined parameter θ. Second, it executed the designed quantum circuit to each element of the batch. Finally, it applied measurements and sent the four dimensional classical vector to a classical fully connected network, converting it into two values: whether an area was prone to a wildfire or not.

F. Libraries

The architecture was built using the Python PennyLane Library with the PyTorch interface with a depth of 6. We used Pennylane's Default Qubit to test the circuit.

G. Hyper-parameters

The hyper-parameters, which are the parameters that define the training process of a neural network, are chosen to maximize accuracy. A batch size of 32 was chosen with a learning rate of 0.0004. For the classical layers, the ReLU activation function is chosen to introduce non-linearity in the network [19] as it shows promise in solving problems like the vanishing gradient problem. The binary cross entropy loss function was used as it was specifically designed for binary classification tasks [19]. The Adam Optimizer was used both for the quantum and classical layers as it is popular with classification tasks [3], and can effectively handle vast amounts of data.

IV. Results and Discussion

A. Performance Metrics

To compare the classical ResNet-18 and our proposed model, the accuracy, recall, precision and F1 scores are used:

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{19}$$

$$recall = \frac{TP}{TP + FN} \tag{20}$$

$$precision = \frac{TP}{TP + FP} \tag{21}$$

$$F_1 = \frac{2 * recall * precision}{recall + precision} \tag{22}$$

B. Training and Validation

In the experiment, both models were trained for a minimum of 20 epochs, with 100 images allocated for testing. Furthermore, the QCNN model was extended to run for an additional ten epochs, totaling 30 epochs. Throughout the training process, the accuracy and loss for both the training and validation datasets were recorded and reported at each epoch.

1) Classical Model: As seen in 3, after 20 epochs, the classical ResNet18 model reported a training accuracy of 81.34% with a loss value of 0.2707. The validation set returned an accuracy of 87.34% with a loss value of 0.2190.

TABLE I: Comparison of classical to hybrid model

Model	Training Accuracy	Validation Accuracy	Training Loss	Validation Loss	Testing Accuracy	Recall	Precision	F1-Score
Hybrid	**0.9340**	**0.9550**	**0.2237**	**0.1953**	**0.9550**	**0.980**	**0.951**	**0.966**
Classical	0.8134	0.8734	0.2707	0.2190	0.8843	0.91	0.892	0.901

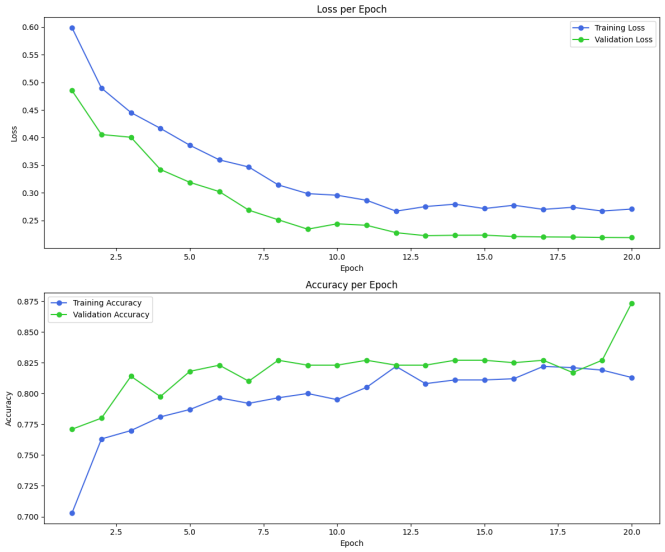


Fig. 3: Loss and Accuracy for Classical Model

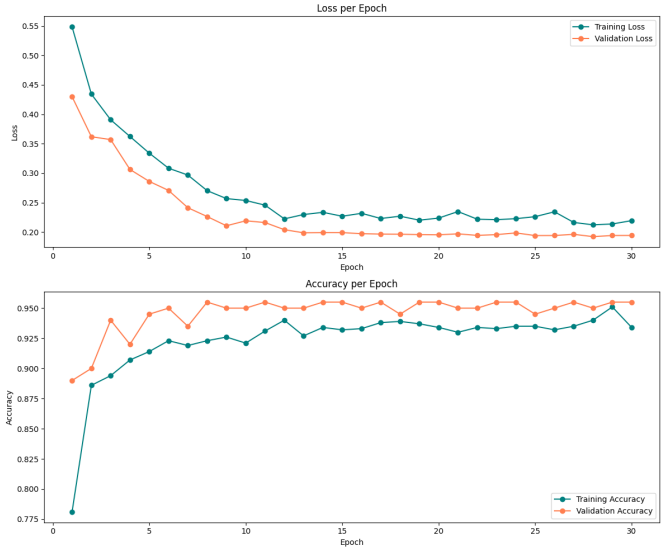


Fig. 4: Loss and Accuracy for Hybrid Model

2) Hybrid Model: After the same number of epochs, the hybrid network had a training accuracy of 93.40% with a loss value of 0.2237, and a validation accuracy of 95.50% with a loss value of 0.1953. After a total of 30 epochs, the hybrid network showed the same loss values and accuracies as it did after 20 epochs, as shown in 4, demonstrating a lack of variability compared to the classical model.

While both models drastically improve in accuracy between epochs 1-3, the hybrid model's accuracy values are consistently higher than the classical model's values, and the hybrid model's loss values are consistently lower than its classical counterpart.

C. Testing

The classical model testing results are shown in the top matrix of 5. Out of the 100 normal images tested, 89 were correctly classified as wildfire prone and 11 were incorrectly classified as not wildfire prone. Additionally, out of the 100 not wildfire prone images tested, 91 were correctly classified as not wildfire prone and 9 were incorrectly classified as wildfire prone. In the bottom matrix of 5, which included the hybrid model testing results, 95 wildfire prone images were correctly classified as wildfire prone and 5 wildfire prone images were incorrectly classified as not wildfire prone. The hybrid model also correctly classified 98 not wildfire prone images and incorrectly classified 2 not wildfire prone images as wildfire prone.

As shown in I, the hybrid model performed better in every metric calculated: accuracy, loss, recall, precision and F1-score. Since it has a higher accuracy and F1-score, it is better at correctly predicting the right label for an image across both classes, wildfire prone and not wildfire prone. Since the hybrid model has a better recall and precision than the classical model, it is both better at predicting correct labels out of the true positives and has a better accuracy for predicting positive labels out of the dataset.

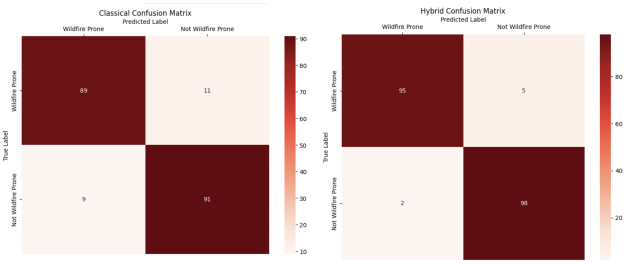


Fig. 5: Confusion Matrices

D. Efficiency

It was observed that the hybrid quantum classical neural network took less time to train. While the standard ResNet18 model took 51 minutes and 25 seconds to train 20 epochs, the hybrid model took 66 minutes and 2 seconds for 30 epochs. This corresponds to a 14.51% increase in speed for the hybrid model over the classical model.

V. Conclusion

This paper acts as a strong proof of concept for the integration of quantum computing into pre-existing models for image

classification tasks. By implementing a variational quantum circuit (VQC) into the ResNet18 network, a higher accuracy in training, validation and testing is achieved alongside higher efficiency. In terms of machine learning approaches to wildfire detection, this paper lays the foundation for subsequent investigations to leverage quantum computing in more complex architectures which incorporate a larger variety of input data, such as climate data [23].

In the context of quantum machine learning, this paper has demonstrated that high fidelity image classification in the *Noisy Intermediate Scale Quantum* era is possible through hybrid models, taking advantage of quantum mechanical properties to perform linear transformations and classical technology to perform non-linear transformations. Future works could explore employing an advanced feature map in the VQCs utilized by the neural network. Additionally, a comprehensive review of VQC architectures that use different embedding and variational layers in hybrid models might yield beneficial outcomes in enhancing results. Finally, investigating the large-scale deployment of hybrid models in a cloud computing environment would be beneficial to real-world applications in the future.

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